

FATHIMA SAFVA

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SUMMARY

Second-year BSc Computer Science student at the University of West London, targeting AI Security and adversarial machine learning. Hands-on experience across six internships spanning backend AI engineering, ML pipeline development, computer vision, and full-stack AI deployment. Built production systems using FastAPI, LLM integration, and real-time AI pipelines. Interested in model robustness, secure AI deployment, LLM safety, and adversarial ML research.

SKILLS

- **Languages & Frameworks:** Python, Java, SQL, JavaScript, HTML5, CSS, C#
- **AI & ML:** Machine Learning, NLP, Computer Vision, TF-IDF, Scikit-learn, TensorFlow, Keras, YOLOv8, ByteTrack, NLTK, OpenCV, Streamlit, face-api.js
- **AI Security & Safety:** Adversarial ML (foundational), LLM Prompt Injection awareness, JWT Authentication, Secure API Design, Rate Limiting, Input Validation
- **AI Tools & Platforms:** Generative AI, Prompt Engineering, LLM Integration (Groq/Llama 3), FastAPI, AIRIA, Google Translate API
- **Data & Analytics:** NumPy, Pandas, Matplotlib, Seaborn, SciPy, Feature Engineering, Statistical Analysis, Data Visualization
- **Development Tools:** GitHub, FastAPI, ASP.NET Core, PostgreSQL, Android Studio, Figma, Canva, VS Code
- **Other:** SEO Optimization, Digital Marketing, Cisco Networking Fundamentals

EXPERIENCE

Backend AI Engineering Intern | FlyRank AI (Remote)

June 2026 – Present

- Built and deployed RESTful APIs using FastAPI with full CRUD operations, database integration, and structured routing for AI-powered backend systems.
- Implemented JWT-based authentication and authorisation to secure API endpoints and manage user access control.
- Developed a polite web scraping pipeline with rate limiting and ethical crawling practices for AI data ingestion workflows.
- Integrated PostgreSQL databases with FastAPI backends for persistent data storage and retrieval in production environments.

Machine Learning Intern | Codveda Technologies (Remote)

May 2026 – June 2026

- Built 6 end-to-end ML pipelines including Linear Regression, KNN, Logistic Regression, K-Means Clustering, Random Forest (GridSearchCV tuning), and a Feed-Forward Neural Network on MNIST using TensorFlow/Keras, achieving 97–100% accuracy across tasks.
- Implemented full model evaluation workflows covering cross-validation, ROC-AUC, confusion matrices, and feature importance analysis using scikit-learn, pandas, matplotlib, and seaborn.
- Documented all projects with portfolio-level GitHub repositories including visualisations, README files, and LinkedIn posts showcasing results and technical insights.

Artificial Intelligence Intern | CodeAlpha (Remote)

March 2026 – April 2026

- Built a real-time Language Translation Tool supporting 25+ languages using Python, Streamlit, and Google Translate API; deployed live on Streamlit Cloud.
- Developed an AI-powered University FAQ Chatbot using TF-IDF vectorization and cosine similarity with NLTK and Scikit-learn.
- Implemented a Real-Time Object Detection and Tracking system using YOLOv8 and ByteTrack with OpenCV, detecting 80+ object classes with persistent tracking IDs.

Data Science Intern | Syntecxhub (Remote)

March 2026 – April 2026

- Delivered 12 end-to-end data science projects across the full pipeline including EDA on Titanic, Netflix, and COVID-19 datasets.
- Built production-quality visualizations using Matplotlib and Seaborn — time series charts, distribution plots, correlation heatmaps, and pairplots.
- Completed a business analytics sprint including retail sales KPI dashboard, IPL performance analysis, and a capstone on global education data across 180 countries.

Machine Learning Intern | Future Interns (Remote)

March 2026 – March 2026

- Built a Sales and Demand Forecasting system using Linear Regression, Random Forest, and Gradient Boosting, achieving 99.2% accuracy with a 6-month revenue forecast.
- Developed an NLP-powered Support Ticket Classification system (TF-IDF + Logistic Regression, 100% accuracy) and an ML-based Resume Screening system using TF-IDF cosine similarity.

Digital Marketing and E-Commerce Assistant | Zenith Groups (Remote)

March 2025 – Sept 2025

- Built and optimized 200+ SEO-focused B2B product listings, improving search visibility and inbound RFQs from global buyers.
- Managed RFQs and coordinated communication with international clients, supporting lead conversion.

- Generated leads and supported learner onboarding through content creation and structured online training sessions across social platforms.

PROJECTS

SOLACE — Emotion-Aware AI Companion | AR26 HackXelerator

April 2026

- Built an emotion-aware AI companion for elderly care and mental wellness, competing in Mission 3 (Human-Robot Interaction) of the AR26 HackXelerator (£10,000 prize pool).
 - Implemented real-time facial emotion detection using face-api.js, conversational AI via Groq/Llama 3, FastAPI backend, and React frontend.
- GitHub: github.com/fsafva13-coder/solace-hri-ar26

Miraé Luxe — Full-Stack Luxury E-Commerce Platform

Feb 2026 - April 2026

- Built a full-stack luxury beauty e-commerce platform with React frontend, ASP.NET Core backend, and PostgreSQL database.
 - Features include an AI beauty assistant, skin quiz, membership system, and order management.
- GitHub: github.com/fsafva13-coder/mirae-luxe

StudyMate AI — AI-Powered Web Application

Jan 2026 – Feb 2026

- Built and deployed an AI-powered study assistant using AIRIA; won 2nd Prize in the AIRIA Track at Decoding Data Science 2026 (13th overall, 576 participants).
 - Designed LLM workflows and prompt logic; integrated AI backend with web frontend and deployed via GitHub Pages.
- GitHub: github.com/fsafva13-coder/studymate-ai

AI Technology Analysis and Autonomous Search-and-Rescue Drone

- Critical analysis of five multimodal AI systems (CLIP, Flamingo, BLIP-2, LLaVA, MiniGPT-4) evaluating architecture, instruction tuning, and ethical risks.
 - Designed an autonomous drone AI system using the PEAS framework, modelling agent behaviour across partially observable, stochastic, and multi-agent disaster environments with sensor fusion and POMDP-style reasoning.
- GitHub: github.com/fsafva13-coder/ai-multimodal-analysis-and-autonomous-drone

EDUCATION

BSc (Hons) Computer Science — Engineering Accredited (IET UK)

Sep 2024 – Present

University of West London

CERTIFICATIONS

- AWS — Fundamentals of Machine Learning and Artificial Intelligence (2026)
- Anthropic — AI Fluency: Framework & Foundations (2026)
- British Airways — Data Science Job Simulation, Forage (2026)
- Top 15 Finalist — Decoding Data Science AI Application Challenge (2026) (2nd Prize, AIRIA Track | Ranked 13th overall among 576 participants)
- One Million Prompters — Dubai Future Foundation (2026)
- Agents and Robotics HackXelerator — KXSB (2026)
- Odoo Technical Workshop — 8H Bootcamp (2026)

References available on request. All projects available on GitHub.